

QF620 Stochastic Modelling in Finance — Session 4

Reference Cheatsheet

Stochastic Integrals and Itô's Formula: stochastic models, the Itô integral, its properties, box calculus, Itô's formula

Built from Session 4 slides (Tee Chyng Wen, SMU) · September 2026

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Source: “Session 4: Stochastic Integrals and Itô's Formula” (Tee Chyng Wen, SMU, 23 slides). Slide numbers are PDF page numbers, shown as [S#]. Material not on the slides is tagged [Beyond slides]. $\triangle!$ marks a slide ambiguity or error, with the corrected version. Prerequisites come from Sessions 1–3 of the same course.

Exam context: the exam format is not stated on the Session 1–4 slides; check the course outline. Whatever the format, from this session you need: the definition of the Itô integral (left endpoint), its five properties and how to use them to get means and variances by hand, the box-calculus table, and both versions of Itô's formula applied to simple functions (X^2 , e^{X+t} , W^2 , W^3 , exponentials). Python here only supports self-checking by simulation.

Notation used throughout (slide conventions)

Symbol	Meaning	Type	Notes
W_t	standard Brownian motion (Wiener process)	random, continuous in t	$W_0 = 0$, $W_t - W_s \sim N(0, t - s)$, independent increments (Session 3)
S_t	stock price process	random	Session 4 model: $S_t = S_0 + \mu t + \sigma W_t$ (arithmetic, not yet GBM)
X_t	a general Itô process	random	$dX_t = \mu_t dt + \sigma_t dW_t$ [S22]
Y_t	a function of a process, $f(X_t)$ or $g(t, X_t)$	random	the output of Itô's formula
$\mu, \mu(t, S_t), \mu_t$	drift coefficient	per unit time	constant [S2], function of (t, S_t) [S3], or adapted process [S22]
$\sigma, \sigma(t, S_t), \sigma_t$	diffusion (volatility) coefficient	per $\sqrt{\text{time}}$	multiplies dW_t ; σ^2 multiplies dt
$\Delta t, \Delta W_t$	finite time step; BM increment $W_{t+\Delta t} - W_t$	—	$\Delta W_t \sim N(0, \Delta t)$
dt, dW_t	infinitesimal limits of $\Delta t, \Delta W_t$	—	shorthand for integral statements; dW_t/dt does <i>not</i> exist [S5]
$P_n = \{t_0, \dots, t_n\}$	partition of $[0, T]$	—	$t_0 = 0, t_n = T$; mesh $\rightarrow 0$ as $n \rightarrow \infty$
x_i	evaluation point in $[t_{i-1}, t_i]$	—	Riemann: anywhere; Itô: must be t_{i-1} [S8]
I_T	$\int_0^T f(u, W_u) dW_u$	random variable	the stochastic integral [S9]
$\mathbb{E}_t[\cdot], \mathbb{E}_s[\cdot]$	conditional expectation given information to time t	—	Session 3 notation for martingales
f', f''	derivatives of $f(x)$	—	needs $f \in C^2$ [S22]
g_t, g_x, g_{xx}	partial derivatives of $g(t, x)$	—	[S23]; no g_{tt} or g_{tx} term survives
$N(m, v)$	normal with mean m , variance v	—	slides write $N(0, \sigma^2 t)$: second argument is a variance
$\mathbb{V}[\cdot]$	variance	—	slides write $V[\cdot]$

Notation trap: on [S10]–[S11], $\Delta W_{t_{i-1}}$ means $W_{t_i} - W_{t_{i-1}}$ (the increment *starting* at t_{i-1}), a forward increment labelled by its left endpoint.

0. Topic map

PREREQUISITES

- S1 Probability: moments, normal & lognormal, MGF $E[e^{\theta X}] = e^{\mu\theta + \sigma^2\theta^2/2}$
- S2 Binomial tree, replication, risk-neutral measure
- S3 Brownian motion: $W_0 = 0$, continuous, indep. stationary increments, $W_t - W_s \sim N(0, t-s)$; $\text{Cov}(W_s, W_t) = s$; nowhere differentiable; Markov; martingales $W_t, W_{t^2-t}, e^{\sigma W_t - \sigma^2 t/2}$; GBM $X_t = e^{W_t}$
 - | "W_t alone is not a good stock model: need drift and volatility" [S2]



SESSION 4: STOCHASTIC INTEGRALS AND ITÔ'S FORMULA

- └ MODELS [S2–S5]
 - | └ $S_t = S_0 + \mu t + \sigma W_t \sim N(S_0 + \mu t, \sigma^2 t)$ [S2]
 - | └ Integral form $\Leftrightarrow$ differential form $dS_t = \mu dt + \sigma dW_t$; $\mu(t, S_t), \sigma(t, S_t)$ [S3]
 - | └ Discrete step: $S_{t+\Delta t} = S_t + \mu \Delta t$ (known at t) + $\sigma \Delta W$ (new randomness) [S4]

└ Can't divide by dt: W is nowhere differentiable $\Rightarrow$ go via integrals	[S5]
STOCHASTIC INTEGRAL [S6–S12]	
└ $S_T = S_0 + \int \mu \, du$ (Riemann) + $\int \sigma \, dW$ (stochastic)	[S6]
└ Riemann: $\sum f(x_i)(t_i - t_{i-1})$, any x_i in the interval	[S6–S7]
└ Itô: $\sum f(t_{i-1})(W_{t_i} - W_{t_{i-1}})$ – LEFT endpoint (previsibility)	[S8]
└ Properties: $E=0$; isometry $E[I^2]=E[f^2]$; deterministic $f \Rightarrow$ normal;	
└ covariance rule; Itô isometry	[S9]
└ Proofs: independence of increments (1); cross terms vanish (2)	[S10–S11]
└ Example: $\int_0^T W \, dW \rightarrow$ mean 0, variance $T^2/2$	[S12]
BOX CALCULUS [S13–S14]	
└ $SLLN \Rightarrow \sum (\Delta W)^2 \approx \sum \Delta t$: $(\Delta W)^2$ is of order Δt	[S13]
└ $dt \cdot dt = 0$, $dt \cdot dW = 0$, $dW \cdot dW = dt \Rightarrow (dX)^2 = \sigma^2 \, dt$	[S14]
ITÔ'S FORMULA [S15–S23]	
└ History: Brown 1827 $\rightarrow$ Einstein/Perrin $\rightarrow$ Wiener 1923 $\rightarrow$ Itô 1942	[S15–S16]
└ Why: chain/product rules without returning to the definition; options are functions of S_t	[S17–S18]
└ Ordinary Taylor: $(\Delta t)^2$ terms vanish $\Rightarrow dx = f'(t) \, dt$	[S19]
└ $f(W)$: $dX = f'(W)dW + \frac{1}{2}f''(W)dt$ ← the "extra term"	[S20]
└ $f(S)$, $dS = \mu dt + \sigma dW$: $dX = [\mu f' + \frac{1}{2}\sigma^2 f'']dt + \sigma f' dW$	[S21]
└ General $f(X_t)$: same with μ_t , σ_t ; Example $Y = X^2$	[S22]
└ $g(t, X_t)$: $dY = [g_t + \mu g_x + \frac{1}{2}\sigma^2 g_{xx}]dt + \sigma g_x dW$; Example $Y = e^{X+t}$	[S23]

LATER (expected, not confirmed by these slides): GBM $dS = \mu S \, dt + \sigma S \, dW$ and its solution via $\ln S$; Black–Scholes PDE via Itô + replication; Girsanov / change of measure; risk-neutral pricing as E^Q of discounted payoffs (ties back to S2).

Dependency logic:

- Session 3 gives the increments $\Delta W \sim N(0, \Delta t)$, independent of the past. Every result this week is built from exactly that fact.
- Independence of ΔW from the (left-endpoint) integrand gives $E[I_T] = 0$; independence across non-overlapping increments gives the isometry.
- $(\Delta W)^2 \approx \Delta t$ (box calculus) is the only reason Itô's formula differs from the ordinary chain rule.
- Itô's formula then lets you (i) evaluate integrals like $\int W \, dW$ without Riemann sums, (ii) check martingales by looking for a zero dt term, (iii) derive SDEs of option prices $V(t, S_t)$.

1 Concept entries

1.1 Building a stock model: Brownian motion with drift [S2–S3]

- **Why not W_t alone** [S2]: we need to (1) control the drift over time and (2) control the volatility.
- **Model** [S2]: $S_t = S_0 + \mu t + \sigma W_t$, with $\mu, \sigma \in \mathbb{R}$.
 - $E[S_t] = S_0 + \mu t$ and $V[S_t] = V[\sigma W_t] = \sigma^2 t$.
 - $S_t \sim N(S_0 + \mu t, \sigma^2 t)$: an affine function of a normal is normal.
- **Two equivalent notations** [S3]:

$$S_t = S_0 + \int_0^t \mu \, du + \int_0^t \sigma \, dW_u \quad \Longleftrightarrow \quad dS_t = \mu \, dt + \sigma \, dW_t.$$

- The differential form is *shorthand* for the integral form. It only has meaning through the integral.
- General case: $dS_t = \mu(t, S_t) \, dt + \sigma(t, S_t) \, dW_t$ (coefficients depend on time and the current state).
- With constant μ, σ the SDE is solved by $S_t = S_0 + \mu t + \sigma W_t$ [S3].
- **Interpretation:** μ sets where the price is expected to go; σ scales how much the path wanders around that trend. Standard deviation grows like $\sqrt{t}$, not t .
- **Weaknesses** [Beyond slides]: S_t is normal, so it can go negative, and the volatility is in price units rather than proportional to the price. This is why the Black–Scholes world uses GBM, $dS = \mu S \, dt + \sigma S \, dW$, which Itô's formula lets us handle (1.13).
- **Worked example** [Beyond slides]: $S_0 = 100$, $\mu = 5/\text{yr}$, $\sigma = 20/\sqrt{\text{yr}}$, $t = 1$. Then $S_1 \sim N(105, 400)$, s.d. 20, and $P(S_1 < 0) = \Phi(-105/20) = \Phi(-5.25) \approx 7.6 \times 10^{-8}$: tiny, but not zero.
- **⚠** [S3] says “a stochastic process is a continuous process that can be written” in this form. Strictly, that describes an *Itô process* (a diffusion). Not every stochastic process has this form (e.g. jump processes).
- **Traps:**
 - Writing $N(S_0 + \mu t, \sigma t)$: the variance is $\sigma^2 t$.

- Treating $dS_t = \mu dt + \sigma dW_t$ as “divide by dt ” algebra.
- **Connections:** S3 (BM properties); S1 (normal distribution); later GBM and Black–Scholes.

1.2 From discrete steps to an SDE; why we cannot differentiate [S4–S5]

- **Discrete update** [S4]:

$$\underbrace{S_{t+\Delta t}}_{\text{future}} = \underbrace{S_t}_{\text{present}} + \underbrace{\mu(t, S_t)\Delta t}_{\text{deterministic}} + \underbrace{\sigma(t, S_t)(W_{t+\Delta t} - W_t)}_{\text{stochastic}}.$$

- **Slide question** [S4]: how is the drift “deterministic” if S_t is random?
 - [Beyond slides] Answer: it is deterministic *conditional on information at time t* . At time t , S_t is already observed, so $\mu(t, S_t)\Delta t$ is a known number over the step $[t, t + \Delta t]$.
 - The only new randomness over the step is $W_{t+\Delta t} - W_t$, which is independent of everything known at t .
 - Viewed from time 0 the drift term is random; viewed from time t it is known. This is exactly the previsibility idea of [S8].
- **Driving noise** [S5]: using BM increments means the price is driven by independent normal shocks.
- **The limit** [S5]: $\Delta t \rightarrow 0^+$ gives the SDE $dS_t = \mu(t, S_t)dt + \sigma(t, S_t)dW_t$.
- **Why not an ODE** [S5]: we would like $\frac{dS_t}{dt} = \mu + \sigma \frac{dW_t}{dt}$, but W is nowhere differentiable, so dW_t/dt does not exist.
 - [Beyond slides] Intuition: $\Delta W/\Delta t$ has standard deviation $\sqrt{\Delta t}/\Delta t = 1/\sqrt{\Delta t} \rightarrow \infty$.
 - Resolution: define everything through integrals (1.3–1.4).
- **Traps:** calling the drift term “non-random” without the conditioning qualifier; writing dW_t/dt anywhere in an answer.

1.3 Integral form and the Riemann integral recap [S6–S7]

- **Integral form** [S6]:

$$S_T = S_0 + \underbrace{\int_0^T \mu(u, S_u) du}_{\text{Riemann integral}} + \underbrace{\int_0^T \sigma(u, S_u) dW_u}_{\text{stochastic integral}}.$$

The first integral is ordinary calculus (path by path); the second is new.

- **Riemann integral** [S6–S7]: for a partition $P_n = \{t_0 = 0, t_1, \dots, t_n = T\}$, f is Riemann integrable if

$$\lim_{n \rightarrow \infty} \sum_{i=1}^n f(x_i)(t_i - t_{i-1}), \quad x_i \in [t_{i-1}, t_i],$$

converges. Picture [S7]: area under the curve, approximated by rectangles (the slide uses 5 intervals on $[a, b]$).

- **Key property:** for a regular f , the limit does *not* depend on where x_i sits in the interval. This fails for stochastic integrals (1.4).
- **Worked example** [Beyond slides]: $\int_0^1 t dt$ with 5 equal intervals.
 - Left points: $(0 + 0.2 + 0.4 + 0.6 + 0.8)(0.2) = 0.40$.
 - Right points: $(0.2 + \dots + 1.0)(0.2) = 0.60$.
 - Both $\rightarrow 0.5$ as $n \rightarrow \infty$; the gap $= 0.2$ shrinks like $1/n$.
- **⚠ Typo** [S7]: “which we would like to integral” should read “integrate”.

1.4 Definition of the stochastic (Itô) integral; previsibility [S8]

- **Definition** [S8]:

$$\int_0^T f(t, W_t) dW_t = \lim_{n \rightarrow \infty} \sum_{i=1}^n f(t_{i-1}, W_{t_{i-1}}) (W_{t_i} - W_{t_{i-1}}).$$

- **Three differences from Riemann** [S8]:
 1. The result is a **random variable** (it depends on the path), not a number.
 2. The integrand **must be evaluated at the left endpoint** t_{i-1} .
 3. **Previsibility:** the value of the integrand is known only at the start of each interval. Choosing $x_i \neq t_{i-1}$ gives *different results*.
- **Finance reading** [Beyond slides]: $f(t_{i-1})$ is a holding (number of shares) chosen at t_{i-1} ; $W_{t_i} - W_{t_{i-1}}$ is the price change that follows. The sum is trading P&L. You cannot choose today's position using tomorrow's price, hence the left endpoint.

- **Why the endpoint matters** [*Beyond slides*] (for $f = W$):

$$\underbrace{\sum W_{t_i} \Delta W_i}_{\text{right}} - \underbrace{\sum W_{t_{i-1}} \Delta W_i}_{\text{left}} = \sum (\Delta W_i)^2 \rightarrow T \neq 0.$$

In Riemann integrals the analogous gap is $\sum \Delta f \Delta t \rightarrow 0$. With BM, $(\Delta W)^2$ is of order Δt , so the gap survives (box calculus, 1.8).

- **Worked example** [*Beyond slides*]: $T = 1, n = 4$, path $W = (0, 0.3, -0.1, 0.4, 0.2)$ at $t = 0, 0.25, 0.5, 0.75, 1$.
 - Increments: $\Delta W = (0.3, -0.4, 0.5, -0.2)$.
 - Left (Itô) sum: $0(0.3) + 0.3(-0.4) + (-0.1)(0.5) + 0.4(-0.2) = -0.12 - 0.05 - 0.08 = -0.25$.
 - Right sum: $0.3(0.3) + (-0.1)(-0.4) + 0.4(0.5) + 0.2(-0.2) = 0.09 + 0.04 + 0.20 - 0.04 = 0.29$.
 - Gap $= 0.54 = \sum (\Delta W)^2 = 0.09 + 0.16 + 0.25 + 0.04 \checkmark$.
 - Check against the exact identity (1.7): left sum $= \frac{1}{2}(W_T^2 - \sum (\Delta W)^2) = \frac{1}{2}(0.04 - 0.54) = -0.25 \checkmark$.
- **⚠** [S8] writes the summand as $f(t_{i-1})$ although the integrand is $f(t, W_t)$; read it as $f(t_{i-1}, W_{t_{i-1}})$ (as on [S10]). “The limit exists in an appropriate sense” means [*Beyond slides*] a mean-square (L^2) limit for square-integrable adapted integrands, not a path-by-path limit.
- **Traps:**
 - Using the midpoint or right endpoint (midpoint gives the Stratonovich integral, 2.4).
 - Expecting a number as the answer to “evaluate $\int_0^T W dW$ ” — it is a random variable, $\frac{1}{2}(W_T^2 - T)$.

1.5 Properties of stochastic integrals [S9]

For $I_T = \int_0^T f(u, W_u) dW_u$ with f adapted and square integrable:

#	Property	Use it for
1	$\mathbb{E}[I_T] = 0$	mean of any Itô integral is zero
2	$\mathbb{E}[I_T^2] = \mathbb{E}\left[\int_0^T f(u, W_u)^2 du\right]$	variance (since the mean is 0); turns a dW integral into a du integral
3	f deterministic $\Rightarrow I_T \sim N\left(0, \int_0^T f(u)^2 du\right)$	full distribution; probabilities
4	$\mathbb{E}\left[\int_0^T f(u) dW_u \int_0^T g(s) dW_s\right] = \int_0^T f(u)g(u) du$	covariances between two integrals (e.g. with $W_T = \int_0^T 1 dW$)
5	Itô's isometry: $\mathbb{E}\left[\left(\int_0^T X_t dW_t\right)^2\right] = \mathbb{E}\left[\int_0^T X_t^2 dt\right]$	same as 2 for a general adapted integrand X_t

- **Recipe for “find mean and variance”:**
 1. Mean $= 0$ (property 1). No computation.
 2. Variance $= \mathbb{E}[I_T^2] = \int_0^T \mathbb{E}[f(u, W_u)^2] du$ (property 2, then swap $\mathbb{E}$ and $\int$ by Fubini).
 3. Compute $\mathbb{E}[f^2]$ with Session 3 moments: $\mathbb{E}[W_u^2] = u$, $\mathbb{E}[W_u^4] = 3u^2$, $\mathbb{E}[e^{2\sigma W_u}] = e^{2\sigma^2 u}$.
- **Worked examples** [*Beyond slides*]:
 - $\int_0^T dW_u = W_T$: variance $\int_0^T 1 du = T \checkmark$ (consistency with BM).
 - $\int_0^1 2u dW_u \sim N(0, \int_0^1 4u^2 du) = N(0, 4/3)$. So $P(I > 1) = 1 - \Phi(1/\sqrt{4/3}) = 1 - \Phi(0.866) \approx 0.193$.
 - $\int_0^T u dW_u \sim N(0, T^3/3)$.
 - $\text{Cov}\left(W_T, \int_0^T u dW_u\right) = \int_0^T 1 \cdot u du = T^2/2$ (property 4 with $f = 1, g = u$).
 - $\mathbb{V}\left[\int_0^T W_u^2 dW_u\right] = \int_0^T \mathbb{E}[W_u^4] du = \int_0^T 3u^2 du = T^3$.
 - $\mathbb{V}\left[\int_0^T e^{W_u} dW_u\right] = \int_0^T \mathbb{E}[e^{2W_u}] du = \int_0^T e^{2u} du = \frac{1}{2}(e^{2T} - 1)$.
- **⚠** [S9] property 4 is stated for deterministic f, g , so the outer $\mathbb{E}$ on the right is redundant (it is just $\int fg du$). The general adapted version is $\mathbb{E}[\int X dW \int Y dW] = \mathbb{E}[\int XY dt]$ [*Beyond slides*].
- **⚠** Properties 2 and 5 are the same statement (5 is the named theorem for a general integrand). Property 3 needs f deterministic: $\int_0^T W dW$ is *not* normal (it is $\frac{1}{2}(W_T^2 - T)$, a shifted scaled χ_1^2).
- **Traps:**
 - Squaring inside instead of outside: $\mathbb{E}[(\int f dW)^2] \neq (\int \mathbb{E}f dW)^2$.
 - Keeping dW after applying isometry: it becomes du .
 - Claiming normality for a random integrand.

1.6 Proofs of properties (1) and (2) [S10–S11]

- **Proof of (1)** [S10]:
 1. Write the integral as the limit of left-point sums.
 2. Take $\mathbb{E}$ inside the (finite) sum.
 3. $f(t_{i-1}, W_{t_{i-1}})$ depends only on the path up to t_{i-1} ; the increment $W_{t_i} - W_{t_{i-1}}$ is independent of it. So $\mathbb{E}[f \cdot \Delta W] = \mathbb{E}[f] \mathbb{E}[\Delta W]$.
 4. $\mathbb{E}[\Delta W] = 0$, so every term vanishes.
- **Proof of (2)** [S11]: write I_T^2 as a double sum over i, j .
 - $i = j$: $\mathbb{E}[f_{i-1}^2 (\Delta W_{i-1})^2] = \mathbb{E}[f_{i-1}^2] \cdot (t_i - t_{i-1})$ (independence, and $\mathbb{E}[(\Delta W)^2] = \Delta t$).
 - $i < j$ (and symmetric): the later increment ΔW_{j-1} is independent of $f_{i-1} f_{j-1} \Delta W_{i-1}$ (all known by t_{j-1}), so the term is $\mathbb{E}[\dots] \cdot 0 = 0$.
 - The diagonal sum is a Riemann sum for $\mathbb{E} \int_0^T f^2 dt$.
- **The one idea behind both:** the integrand is fixed before the increment happens. Using the right endpoint breaks both proofs because $f(t_i, W_{t_i})$ and ΔW_i are correlated.
- **⚠ [S10]** exchanges $\mathbb{E}$ and $\lim$ without comment (justified by L^2 convergence [Beyond slides]) and has a typo in the third line's summation index (" $i = i$ " should be " $i = 1$ ").

1.7 Worked example: $\int_0^T W_t dW_t$ [S12]

- **Slide question** [S12]: find $\mathbb{E}[I_T]$ and $\mathbb{V}[I_T]$ for $I_T = \int_0^T W_t dW_t$.
- **Solution with the properties:**
 - $\mathbb{E}[I_T] = 0$ (property 1).
 - $\mathbb{V}[I_T] = \mathbb{E}[I_T^2] = \mathbb{E} \int_0^T W_t^2 dt = \int_0^T t dt = \frac{T^2}{2}$ (property 2).
- **Closed form** [Beyond slides] (via Itô, 1.11): $d(W_t^2) = 2W_t dW_t + dt$, so

$$\int_0^T W_t dW_t = \frac{1}{2} W_T^2 - \frac{1}{2} T.$$

- Check the mean: $\frac{1}{2} \mathbb{E}[W_T^2] - \frac{1}{2} T = \frac{1}{2} T - \frac{1}{2} T = 0 \checkmark$.
- Check the variance: $\frac{1}{4} \mathbb{V}[W_T^2] = \frac{1}{4} (\mathbb{E}[W_T^4] - T^2) = \frac{1}{4} (3T^2 - T^2) = T^2/2 \checkmark$.
- Ordinary calculus would say $\int W dW = \frac{1}{2} W_T^2$; the $-\frac{1}{2} T$ is the Itô correction.
- **Direct derivation** [Beyond slides] (no Itô's formula): with $\Delta W_i = W_{t_i} - W_{t_{i-1}}$,

$$W_{t_{i-1}} \Delta W_i = \frac{1}{2} (W_{t_i}^2 - W_{t_{i-1}}^2) - \frac{1}{2} (\Delta W_i)^2 \Rightarrow \sum_i W_{t_{i-1}} \Delta W_i = \frac{1}{2} W_T^2 - \frac{1}{2} \sum_i (\Delta W_i)^2 \rightarrow \frac{1}{2} W_T^2 - \frac{1}{2} T.$$

- **Implementation (tested):**

```
import numpy as np

def brownian_paths(n_paths, n_steps, T, seed=0):
    """Simulate W on a uniform grid: t (n_steps+1), W (n_paths, n_steps+1)."""
    rng = np.random.default_rng(seed)
    dt = T / n_steps
    dW = rng.normal(0.0, np.sqrt(dt), size=(n_paths, n_steps))
    W = np.concatenate([np.zeros((n_paths, 1)), dW.cumsum(axis=1)], axis=1)
    return np.linspace(0, T, n_steps + 1), W

def ito_integral(f_vals, W):
    """Left-point (Ito) sum sum_i f(t_{i-1}) (W_{t_i} - W_{t_{i-1}}), per path."""
    return (f_vals[:, :-1] * np.diff(W, axis=1)).sum(axis=1)

t, W = brownian_paths(100_000, 500, T=1.0)
I = ito_integral(W, W) # int_0^1 W dW
print(I.mean(), I.var()) # -0.001, 0.494 (theory 0, T^2/2 = 0.5)
```

Pitfall: the simulated I differs from $\frac{1}{2}(W_T^2 - T)$ path by path by exactly $\frac{1}{2}(T - \sum (\Delta W)^2)$ (up to 0.15 here with 500 steps). That is discretisation, not a bug; it shrinks as the grid is refined.

- **Traps:** answering $\mathbb{V} = T$ (confusing with $\mathbb{V}[W_T]$); answering $\mathbb{V} = T^2$ (forgetting the $\frac{1}{2}$ from $\int t dt$).

1.8 Box calculus: $(dW)^2 = dt$ [S13–S14]

- **Strong law of large numbers** [S13]: the average of i.i.d. random variables converges to their mean with probability 1. Example: i.i.d. ± 1 coin flips average to 0.
- **The argument** [S13–S14]:
 - Split $[0, T]$ into n steps of $\Delta t = T/n$. Each $(\Delta W_i)^2$ has mean Δt and variance $2\Delta t^2$.
 - $\sum_i (\Delta W_i)^2 = T \cdot \frac{1}{n} \sum_i \frac{(\Delta W_i)^2}{\Delta t} \rightarrow T \cdot 1 = T$ by the SLLN (each $(\Delta W_i)^2/\Delta t \sim \chi_1^2$ has mean 1).
 - So in aggregate $(\Delta W)^2$ behaves like Δt : it is of order Δt , hence ΔW is of order $\sqrt{\Delta t}$.
- **Rules** [S14] ($\Delta \rightarrow d$ as $\Delta t \rightarrow 0$):

$\cdot$	dt	dW_t
dt	0	0
dW_t	0	dt

- $(dt)^2 = 0$: order Δt^2 , negligible.
- $dt \cdot dW_t = 0$: order $\Delta t^{3/2}$, negligible.
- $(dW_t)^2 = dt$: order Δt , survives.
- **Workhorse corollary** [S14]: $dX_t = \mu dt + \sigma dW_t \Rightarrow (dX_t)^2 = \sigma^2 dt$ (only the $dW \cdot dW$ term survives).
- *[Beyond slides]* Extensions you will meet: $(dW)^3 = 0$; for two *independent* BMs $dW^1 dW^2 = 0$; for correlated BMs $dW^1 dW^2 = \rho dt$. Products of two Itô processes: $dX dY = \sigma_X \sigma_Y dt$.
- **Numerical check** *[Beyond slides]* (2,000 paths, $T = 1$): mean and s.d. of $\sum (\Delta W)^2$ are 0.988 ± 0.446 ($n = 10$), 0.998 ± 0.141 ($n = 100$), 0.998 ± 0.045 ($n = 1000$), 1.000 ± 0.014 ($n = 10,000$). The s.d. is $\sqrt{2/n}T$: the sum collapses onto T .
- **!** [S13] writes the SLLN as “ $n \rightarrow \infty \Rightarrow P(\frac{X_1 + \dots + X_n}{n} = \mu) = 1$ ”. The correct statement is $P(\lim_{n \rightarrow \infty} \frac{X_1 + \dots + X_n}{n} = \mu) = 1$ (the limit is inside the probability).
- **Traps:**
 - Writing $(dX)^2 = (\mu^2 + \sigma^2) dt$ — the $\mu^2 (dt)^2$ term is zero.
 - Reading $(dW)^2 = dt$ as “ $(\Delta W)^2$ equals Δt for each step”. A single squared increment is random; only the sum converges.

1.9 History [S15–S16]

- 1827: Robert Brown (botanist) observes pollen particles moving in water.
- Early 1900s: Einstein, Perrin and other physicists study Brownian motion (Bachelier's 1900 thesis already used it for prices; Session 3).
- 1923: Wiener defines probability measures on path spaces, using Lebesgue integration, and lays the foundations of stochastic analysis.
- 1942: Itô rebuilds stochastic integration from scratch and creates the theory of stochastic differential equations.

S16 : portrait and quote of Kiyosi Itô (1915–2008). Not examinable content.

1.10 Why Itô's formula; the ordinary Taylor expansion [S17–S19]

- **Theory** [S17]: once the integral is defined, we want rules (chain rule, product rule) to manipulate it without returning to Riemann-type sums.
- **Practice** [S17]: an option price is a function of the random stock price, $V(t, S_t)$. We need the SDE of V to understand how it evolves (the basis of hedging and Black–Scholes *[Beyond slides]*).

S18 : quotes (Hamming; Föllmer): Itô calculus transformed quantitative finance conceptually and computationally.

- **Ordinary Taylor** [S19]: for $x = f(t)$,

$$\Delta x = f'(t)\Delta t + \frac{1}{2!}f''(t)(\Delta t)^2 + \frac{1}{3!}f'''(t)(\Delta t)^3 + \dots \xrightarrow{\Delta t \rightarrow 0^+} \frac{dx}{dt} = f'(t).$$

All terms from $(\Delta t)^2$ onward are negligible. The whole point of 1.11 is that with ΔW in place of Δt , the second-order term is *not* negligible.

1.11 Itô's formula for $f(W_t)$ [S20]

- **Derivation** [S20]: for $X_t = f(W_t)$,

$$\Delta X_t = f'(W_t)\Delta W_t + \frac{1}{2}f''(W_t)(\Delta W_t)^2 + \dots$$

and with $(dW)^2 = dt$ (and higher powers = 0):

$$dX_t = f'(W_t) dW_t + \underbrace{\frac{1}{2}f''(W_t) dt}_{\text{extra term}}$$

- **Worked examples** [Beyond slides]:

$f(W)$	f'	f''	$df(W_t)$
W^2	$2W$	2	$2W_t dW_t + dt$
W^3	$3W^2$	$6W$	$3W_t^2 dW_t + 3W_t dt$
W^4	$4W^3$	$12W^2$	$4W_t^3 dW_t + 6W_t^2 dt$
e^W	e^W	e^W	$e^{W_t} dW_t + \frac{1}{2}e^{W_t} dt$
$e^{\sigma W}$	$\sigma e^{\sigma W}$	$\sigma^2 e^{\sigma W}$	$\sigma e^{\sigma W_t} dW_t + \frac{1}{2}\sigma^2 e^{\sigma W_t} dt$

- **Martingale test** [Beyond slides]: an Itô process is a (local) martingale iff its dt term is zero. This recovers Session 3 quickly:
 - W^2 has drift dt , so W_t^2 is not a martingale, but $W_t^2 - t$ has $d(W^2 - t) = 2W dW$: martingale ✓ (S3).
 - W^3 has drift $3W dt \neq 0$: not a martingale. Consistent with S3's $\mathbb{E}_s[W_t^3] = W_s^3 + 3(t-s)W_s$ ✓.
- **Using it to compute expectations** [Beyond slides]: integrate and take $\mathbb{E}$ (the dW part has mean 0):

$$\mathbb{E}[W_T^4] = \mathbb{E} \int_0^T 6W_t^2 dt = \int_0^T 6t dt = 3T^2 \quad \checkmark, \quad \mathbb{E}[e^{W_T}] = 1 + \frac{1}{2} \int_0^T \mathbb{E}[e^{W_t}] dt \Rightarrow \mathbb{E}[e^{W_T}] = e^{T/2} \quad \checkmark.$$

- **Traps:** forgetting the $\frac{1}{2}$; putting f'' on dW ; writing $d(W^2) = 2W dW$ (ordinary chain rule).

1.12 Itô's formula for a function of a stochastic process [S21–S22]

- **Derivation** [S21]: with $\Delta S_t = \mu \Delta t + \sigma \Delta W_t$ and $X_t = f(S_t)$,
 - $(\Delta S_t)^2 = \mu^2 (\Delta t)^2 + 2\mu\sigma \Delta t \Delta W_t + \sigma^2 (\Delta W_t)^2 \rightarrow \sigma^2 dt$.
 - $dX_t = f'(S_t) dS_t + \frac{1}{2}f''(S_t)\sigma^2 dt = \left[\mu f'(S_t) + \frac{\sigma^2}{2}f''(S_t) \right] dt + \sigma f'(S_t) dW_t$.
- **General statement** [S22]: if $dX_t = \mu_t dt + \sigma_t dW_t$ and f, f', f'' are continuous, then $Y_t = f(X_t)$ satisfies

$$dY_t = \left[\mu_t f'(X_t) + \frac{1}{2}\sigma_t^2 f''(X_t) \right] dt + \sigma_t f'(X_t) dW_t$$

- **Memory aid** [Beyond slides]: $df = f' dX + \frac{1}{2}f''(dX)^2$, then substitute dX and $(dX)^2 = \sigma^2 dt$. This two-step version avoids memorising the expanded form.
- **Slide example** [S22]: $dX_t = \mu dt + \sigma dW_t$, $Y_t = X_t^2$.
 - $f' = 2X$, $f'' = 2$.
 - $dY_t = 2X_t dX_t + \frac{1}{2} \cdot 2 \cdot \sigma^2 dt = (2\mu X_t + \sigma^2) dt + 2\sigma X_t dW_t$.
- **Numbers** [Beyond slides]: $\mu = 0.1$, $\sigma = 0.2$, $X_0 = 1$. At $t = 0$: drift $2(0.1)(1) + 0.04 = 0.24$, diffusion $2(0.2)(1) = 0.4$. The ordinary chain rule would give drift 0.20; the Itô term adds $\sigma^2 = 0.04$.
- **Consequence** [Beyond slides]: $\frac{d}{dt}\mathbb{E}[Y_t] = 2\mu\mathbb{E}[X_t] + \sigma^2 = 2\mu(X_0 + \mu t) + \sigma^2$, so $\mathbb{E}[X_T^2] = (X_0 + \mu T)^2 + \sigma^2 T$ ✓ (mean² + variance). With the numbers above and $T = 1$: $1.21 + 0.04 = 1.25$; simulation gives 1.2508.
- **Traps:**
 - Multiplying the Itô term by μ or forgetting σ^2 .
 - Diffusion term is $\sigma f'$, not $\sigma f''$ and not $\sigma^2 f'$.

1.13 Itô's formula with time dependence, $g(t, X_t)$ [S23]

- **Statement** [S23]: if $dX_t = \mu_t dt + \sigma_t dW_t$ and g, g_t, g_x, g_{xx} are continuous, then $Y_t = g(t, X_t)$ satisfies

$$dY_t = \left[g_t + \mu_t g_x + \frac{1}{2}\sigma_t^2 g_{xx} \right] dt + \sigma_t g_x dW_t$$

(all partials evaluated at (t, X_t)).

- **Why no g_{tt} or g_{tx}** [Beyond slides]: second-order Taylor terms carry $(dt)^2 = 0$ and $dt dX = 0$. Only $\frac{1}{2}g_{xx}(dX)^2$ survives.
- **Slide example** [S23]: $dX_t = \mu dt + \sigma dW_t$, $Y_t = e^{X_t + t}$.

- $g_t = g_x = g_{xx} = e^{x+t} = Y$.
- $dY_t = (1 + \mu + \frac{1}{2}\sigma^2) Y_t dt + \sigma Y_t dW_t$.
- Proportional drift and volatility: Y is a geometric Brownian motion.
- **Key application: the exponential martingale** [*Beyond slides*] (S3 result, now in one line). $g(t, w) = e^{\sigma w - \frac{1}{2}\sigma^2 t}$ with $X = W$ ($\mu = 0, \sigma_t = 1$):

$$g_t = -\frac{1}{2}\sigma^2 g, \quad g_x = \sigma g, \quad g_{xx} = \sigma^2 g \Rightarrow dg = \left(-\frac{1}{2}\sigma^2 + \frac{1}{2}\sigma^2\right) g dt + \sigma g dW = \sigma g dW_t.$$

No drift, so it is a martingale ✓.

- **Key application: GBM** [*Beyond slides*]. $S_t = S_0 e^{(\mu - \frac{1}{2}\sigma^2)t + \sigma W_t}$ gives $g_t = (\mu - \frac{1}{2}\sigma^2)S$, $g_x = \sigma S$, $g_{xx} = \sigma^2 S$, so

$$dS_t = \mu S_t dt + \sigma S_t dW_t.$$

Conversely, if $dS = \mu S dt + \sigma S dW$, then $d \ln S_t = (\mu - \frac{1}{2}\sigma^2) dt + \sigma dW_t$ (use $f = \ln$, $f' = 1/S$, $f'' = -1/S^2$). The $-\frac{1}{2}\sigma^2$ is the “volatility drag” between arithmetic and log returns.

- **Traps:**
 - Forgetting g_t (the “Session 4 [S22] formula” has no time term).
 - In GBM, dropping the $-\frac{1}{2}\sigma^2$ in the exponent: $S_0 e^{\mu t + \sigma W_t}$ has drift $\mu + \frac{1}{2}\sigma^2$, not μ .

1.14 Simulation toolkit (self-checking) [*Beyond slides*]

- **Evaluation point matters** (100,000 paths, 500 steps, $T = 1$, reusing `brownian_paths` from 1.7):

```
dW = np.diff(W, axis=1)
left = (W[:, :-1] * dW).sum(1) # Ito
right = (W[:, 1:] * dW).sum(1) # right endpoint
mid = (0.5 * (W[:, :-1] + W[:, 1:]) * dW).sum(1) # Stratonovich
print(left.mean(), right.mean(), mid.mean()) # -0.001 0.999 0.499
```

Theory: $\mathbb{E} = 0, T, T/2$ respectively (2.4).

- **Checking the properties:**

```
J = ito_integral(np.broadcast_to(2 * t, W.shape), W) # int_0^1 2t dW
print(J.var()) # 1.327 (theory 4/3)
K = ito_integral(np.broadcast_to(t, W.shape), W) # int_0^1 t dW
print(np.cov(W[:, -1], K)[0, 1]) # 0.498 (theory 1/2)
```

- **Checking Itô's formula** for $Y = X^2$ (1.12):

```
mu, sig, X0, T = 0.1, 0.2, 1.0, 1.0
X = X0 + mu * t + sig * W # exact solution of dX = mu dt + sig dW
print((X[:, -1] ** 2).mean()) # 1.2508 (theory (X0+mu T)^2 + sig^2 T = 1.25)
```

Pitfall: never evaluate an integrand at `W[:, 1:]` inside `ito_integral`: that is the right-endpoint sum and gives $\mathbb{E}[W dW] \approx T$, not 0.

2 Comparison tables

2.1 Riemann vs Itô integral

	Riemann $\int f(t) dt$ [S6–S7]	Itô $\int f(t, W_t) dW_t$ [S8]
Integrator	time increments $t_i - t_{i-1}$ (deterministic)	BM increments $W_{t_i} - W_{t_{i-1}}$ (random)
Evaluation point	any $x_i \in [t_{i-1}, t_i]$	left endpoint t_{i-1} only
Does the point matter?	no (for regular f)	yes: gap $\rightarrow \sum (\Delta W)^2 \rightarrow T$
Result	a number	a random variable
Mean	the value itself	0
Size of a term	order Δt	order $\sqrt{\Delta t}$
Chain rule	ordinary: $df = f' dx$	Itô: $df = f' dX + \frac{1}{2} f'' (dX)^2$
Example	$\int_0^T t dt = T^2/2$	$\int_0^T W dW = \frac{1}{2} W_T^2 - \frac{1}{2} T$

2.2 Ordinary vs Itô calculus

	Ordinary [S19]	Itô [S20–S23]
Taylor terms kept	first order only	first order + $\frac{1}{2}$ × second order in dW
Why	$(\Delta t)^2 \ll \Delta t$	$(\Delta W)^2 \sim \Delta t$
$d(x^2)$	$2x dx$	$2X dX + \sigma^2 dt$
$d(e^x)$	$e^x dx$	$e^X dX + \frac{1}{2}\sigma^2 e^X dt$
$d(\ln x)$	dx/x	$dX/X - \frac{1}{2}\sigma^2/X^2 dt$ (with dX volatility σ)
Product rule	$d(xy) = x dy + y dx$	$d(XY) = X dY + Y dX + dX dY$ [Beyond slides]

2.3 Box calculus table [S14]

$\cdot$	dt	dW_t	order of the product
dt	0	0	$\Delta t^2, \Delta t^{3/2}$: vanish
dW_t	0	dt	Δt : survives

Corollary: $(\mu dt + \sigma dW)^2 = \sigma^2 dt$; $(\mu dt + \sigma dW)(dt) = 0$.

2.4 Evaluation point and the value of $\int_0^T W dW$ [Beyond slides]

Point x_i	Name	Limit of $\sum W_{x_i} \Delta W_i$	Mean
t_{i-1} (left)	Itô (used in this course)	$\frac{1}{2}W_T^2 - \frac{1}{2}T$	0
midpoint / average	Stratonovich	$\frac{1}{2}W_T^2$ (ordinary calculus)	$T/2$
t_i (right)	backward	$\frac{1}{2}W_T^2 + \frac{1}{2}T$	T

Only the left-point version is non-anticipating (previsible), so only it has mean zero and represents trading gains.

2.5 Versions of Itô's formula

Version	Input	dY	Slide example
$f(W_t)$ [S20]	dW	$f' dW + \frac{1}{2}f'' dt$	—
$f(S_t)$ [S21]	$dS = \mu dt + \sigma dW$	$[\mu f' + \frac{1}{2}\sigma^2 f''] dt + \sigma f' dW$	—
$f(X_t)$ [S22]	$dX = \mu_t dt + \sigma_t dW$	$[\mu_t f' + \frac{1}{2}\sigma_t^2 f''] dt + \sigma_t f' dW$	$Y = X^2$
$g(t, X_t)$ [S23]	same	$[g_t + \mu_t g_x + \frac{1}{2}\sigma_t^2 g_{xx}] dt + \sigma_t g_x dW$	$Y = e^{X+t}$

Each row is a special case of the last: $f(W)$ is g with no t and $\mu = 0, \sigma = 1$.

3 Master formula sheet

Models [S2–S5]

- $S_t = S_0 + \mu t + \sigma W_t \sim N(S_0 + \mu t, \sigma^2 t)$.
- $dS_t = \mu(t, S_t) dt + \sigma(t, S_t) dW_t \iff S_T = S_0 + \int_0^T \mu du + \int_0^T \sigma dW_u$.

Integrals [S6–S8]

- Riemann: $\lim \sum f(x_i)(t_i - t_{i-1}), x_i \in [t_{i-1}, t_i]$.

- Itô: $\lim \sum f(t_{i-1}, W_{t_{i-1}})(W_{t_i} - W_{t_{i-1}})$ — left endpoint.

Properties [S9] ($I_T = \int_0^T f \, dW$)

- $\mathbb{E}[I_T] = 0$; $\mathbb{E}[I_T^2] = \mathbb{E} \int_0^T f^2 \, du$ (isometry).
- f deterministic: $I_T \sim N(0, \int_0^T f^2 \, du)$.
- $\mathbb{E}[\int f \, dW \int g \, dW] = \int fg \, du$.
- $\int_0^T W \, dW$: mean 0, variance $T^2/2$; equals $\frac{1}{2}W_T^2 - \frac{1}{2}T$ [Beyond slides].

Box calculus [S13–S14]

- $dt^2 = 0$, $dt \, dW = 0$, $dW^2 = dt$; $(dX)^2 = \sigma^2 dt$.
- $\sum (\Delta W_i)^2 \rightarrow T$ (quadratic variation).

Itô's formula [S20–S23]

- $df(W_t) = f'(W_t) \, dW_t + \frac{1}{2}f''(W_t) \, dt$.
- $df(X_t) = [\mu_t f' + \frac{1}{2}\sigma_t^2 f''] \, dt + \sigma_t f' \, dW_t$.
- $dg(t, X_t) = [g_t + \mu_t g_x + \frac{1}{2}\sigma_t^2 g_{xx}] \, dt + \sigma_t g_x \, dW_t$.
- Shortcut: $dg = g_t \, dt + g_x \, dX + \frac{1}{2}g_{xx} \, (dX)^2$.

Standard results [Beyond slides]

- $d(W^2) = 2W \, dW + dt$; $d(W^3) = 3W^2 \, dW + 3W \, dt$.
- $d(e^{\sigma W_t - \frac{1}{2}\sigma^2 t}) = \sigma e^{\sigma W_t - \frac{1}{2}\sigma^2 t} \, dW_t$ (martingale).
- GBM: $S_t = S_0 e^{(\mu - \frac{1}{2}\sigma^2)t + \sigma W_t} \iff dS = \mu S \, dt + \sigma S \, dW$; $d \ln S = (\mu - \frac{1}{2}\sigma^2) \, dt + \sigma \, dW$.
- Zero dt term $\iff$ (local) martingale.

Moments to have ready (Session 3)

- $\mathbb{E}[W_t^2] = t$, $\mathbb{E}[W_t^3] = 0$, $\mathbb{E}[W_t^4] = 3t^2$, $\mathbb{E}[e^{\theta W_t}] = e^{\frac{1}{2}\theta^2 t}$, $\text{Cov}(W_s, W_t) = \min(s, t)$.

4 Glossary

- **Adapted / non-anticipating**: a process whose value at t uses only information up to t .
- **Box calculus**: multiplication rules $dt^2 = 0$, $dt \, dW = 0$, $dW^2 = dt$ [S14].
- **Brownian motion** W_t : $W_0 = 0$, continuous, independent stationary increments, $W_t - W_s \sim N(0, t - s)$ (S3).
- **Diffusion coefficient** σ : the coefficient on dW_t .
- **Drift coefficient** μ : the coefficient on dt .
- **Geometric Brownian motion (GBM)**: $dS = \mu S \, dt + \sigma S \, dW$; log-normal prices [Beyond slides].
- **Itô integral**: limit of left-endpoint sums $\sum f(t_{i-1})\Delta W_i$ [S8].
- **Itô's formula / lemma**: the stochastic chain rule, with the extra $\frac{1}{2}f''\sigma^2 \, dt$ term [S20–S23].
- **Itô's isometry**: $\mathbb{E}[(\int X \, dW)^2] = \mathbb{E}[\int X^2 \, dt]$ [S9].
- **Itô process**: $X_t = X_0 + \int \mu_s \, ds + \int \sigma_s \, dW_s$, i.e. $dX = \mu \, dt + \sigma \, dW$.
- **Martingale**: $\mathbb{E}_t[M_T] = M_t$ for $T \geq t$ (S3); for Itô processes, zero drift.
- **Nowhere differentiable**: dW/dt does not exist at any t ; why SDEs are written as integrals [S5].
- **Partition**: $0 = t_0 < t_1 < \dots < t_n = T$ [S6].
- **Previsibility**: the integrand's value is fixed at the start of each interval [S8].
- **Quadratic variation**: $\lim \sum (\Delta W_i)^2 = T$ for BM; zero for smooth functions.
- **Riemann integral**: limit of $\sum f(x_i)\Delta t_i$; area under a curve [S6–S7].
- **SDE**: stochastic differential equation, $dS = \mu(t, S) \, dt + \sigma(t, S) \, dW$ [S3, S5].
- **Stratonovich integral**: midpoint version; obeys ordinary calculus but anticipates [Beyond slides].
- **Strong law of large numbers**: sample averages of i.i.d. variables converge to the mean with probability 1 [S13].

5 Self-test

Questions

Q1 (conceptual, easy). $S_t = S_0 + \mu t + \sigma W_t$. State the distribution of S_t , and explain in what sense the term $\mu(t, S_t)\Delta t$ on [S4] is “deterministic”.

Q2 (conceptual, easy). Why can we not write $\frac{dS_t}{dt} = \mu + \sigma \frac{dW_t}{dt}$? What do we do instead?

Q3 (conceptual). Give three differences between a Riemann integral and the Itô integral, and explain previsibility using a trading interpretation.

Q4 (numerical). On $[0, 1]$ with $n = 4$, a path takes values $W = (0, 0.3, -0.1, 0.4, 0.2)$. Compute the left-point and right-point sums approximating $\int_0^1 W \, dW$, and explain their difference.

Q5 (numerical). Find the mean and variance of (a) $\int_0^T W_t \, dW_t$; (b) $\int_0^T t \, dW_t$; (c) $\int_0^T W_t^2 \, dW_t$. Which are normally distributed?

Q6 (numerical). Let $I = \int_0^1 2t \, dW_t$. Find $P(I > 1)$ and $\text{Cov}(I, W_1)$.

Q7 (conceptual). Using the SLLN, argue that $\sum_{i=1}^n (\Delta W_i)^2 \rightarrow T$. Deduce the box-calculus table and $(dX)^2$ for $dX = \mu \, dt + \sigma \, dW$.

Q8 (derivation). Use Itô's formula on $f(W) = W^2$ to find $\int_0^T W \, dW$ in closed form, and verify its mean and variance match Q5(a).

Q9 (slide example, [S22]). $dX_t = \mu \, dt + \sigma \, dW_t$, $Y_t = X_t^2$. Derive dY_t . Using it, find $\mathbb{E}[X_T^2]$ and check against the distribution of X_T .

Q10 (slide example, [S23]). $dX_t = \mu \, dt + \sigma \, dW_t$, $Y_t = e^{X_t+t}$. Derive dY_t . What kind of process is Y ?

Q11 (martingales). Using Itô's formula, decide which are martingales: (a) $W_t^2 - t$; (b) W_t^3 ; (c) $W_t^3 - 3tW_t$; (d) $e^{\sigma W_t - \frac{1}{2}\sigma^2 t}$.

Q12 (hard, synthesis). (a) Show that $S_t = S_0 \exp((\mu - \frac{1}{2}\sigma^2)t + \sigma W_t)$ solves $dS = \mu S \, dt + \sigma S \, dW$. (b) Derive $d \ln S_t$ directly from the SDE. (c) Use Itô's formula on W^4 to show $\mathbb{E}[W_T^4] = 3T^2$. (d) Why must the Itô integrand be evaluated at the left endpoint for property (1) to hold?

Solutions

S1.

- $S_t \sim N(S_0 + \mu t, \sigma^2 t)$: $\mathbb{E} = S_0 + \mu t$, $\mathbb{V} = \sigma^2 t$.
- The drift term is known *at time t*: S_t has been observed, so $\mu(t, S_t)\Delta t$ is fixed for the coming step. The only fresh randomness is $W_{t+\Delta t} - W_t$, independent of the past.

S2.

- W is nowhere differentiable; $\Delta W / \Delta t$ has s.d. $1/\sqrt{\Delta t} \rightarrow \infty$.
- Instead write the integral form $S_T = S_0 + \int \mu \, du + \int \sigma \, dW_u$ and define the stochastic integral as a limit of sums. $dS = \dots$ is shorthand for this.

S3.

- (i) Result is a random variable, not a number. (ii) Integrator is a BM increment, not Δt . (iii) Integrand must be at the left endpoint; other points give different limits.
- Trading: $f(t_{i-1})$ is the position chosen at t_{i-1} ; ΔW_i is the next price move. You cannot pick a position with knowledge of the move it will earn.

S4.

- $\Delta W = (0.3, -0.4, 0.5, -0.2)$.
- Left: $0 - 0.12 - 0.05 - 0.08 = -0.25$. Right: $0.09 + 0.04 + 0.20 - 0.04 = 0.29$.
- Difference = $\sum (\Delta W)^2 = 0.54$, which does not vanish as $n \rightarrow \infty$ (it tends to $T = 1$). So unlike Riemann, the evaluation point matters.

S5.

- All means are 0 (property 1).
- (a) $\mathbb{V} = \int_0^T \mathbb{E}[W_t^2] \, dt = \int_0^T t \, dt = T^2/2$.
- (b) $\mathbb{V} = \int_0^T t^2 \, dt = T^3/3$.
- (c) $\mathbb{V} = \int_0^T \mathbb{E}[W_t^4] \, dt = \int_0^T 3t^2 \, dt = T^3$.
- Only (b) is normal: its integrand is deterministic (property 3). (a) equals $\frac{1}{2}(W_T^2 - T)$, a shifted χ^2 .

S6.

- $I \sim N(0, \int_0^1 4t^2 \, dt) = N(0, 4/3)$, s.d. 1.1547.
- $P(I > 1) = 1 - \Phi(1/1.1547) = 1 - \Phi(0.866) \approx 0.193$.
- $\text{Cov}(I, W_1) = \mathbb{E}[\int_0^1 2t \, dW \int_0^1 1 \, dW] = \int_0^1 2t \, dt = 1$.

S7.

- $(\Delta W_i)^2/\Delta t$ are i.i.d. χ_1^2 with mean 1. By the SLLN, $\frac{1}{n} \sum (\Delta W_i)^2/\Delta t \rightarrow 1$, so $\sum (\Delta W_i)^2 = T \cdot \frac{1}{n} \sum (\Delta W_i)^2/\Delta t \rightarrow T = \sum \Delta t$.
- So $(dW)^2 = dt$. Higher orders vanish: $(dt)^2 \sim \Delta t^2$, $dt dW \sim \Delta t^{3/2}$, both $o(\Delta t)$.
- $(dX)^2 = \mu^2 dt^2 + 2\mu\sigma dt dW + \sigma^2 dW^2 = \sigma^2 dt$.

S8.

- $d(W^2) = 2W dW + \frac{1}{2} \cdot 2 dt = 2W dW + dt$. Integrate: $W_T^2 = 2 \int_0^T W dW + T$, so $\int_0^T W dW = \frac{1}{2} W_T^2 - \frac{1}{2} T$.
- Mean: $\frac{1}{2} T - \frac{1}{2} T = 0 \checkmark$. Variance: $\frac{1}{4} (\mathbb{E} W_T^4 - (\mathbb{E} W_T^2)^2) = \frac{1}{4} (3T^2 - T^2) = T^2/2 \checkmark$.

S9.

- $f' = 2X$, $f'' = 2$: $dY = (2\mu X_t + \sigma^2) dt + 2\sigma X_t dW_t$.
- Take expectations of the integrated form (the dW integral has mean 0): $\frac{d}{dt} \mathbb{E}[Y_t] = 2\mu \mathbb{E}[X_t] + \sigma^2 = 2\mu(X_0 + \mu t) + \sigma^2$.
- Integrate: $\mathbb{E}[X_T^2] = X_0^2 + 2\mu X_0 T + \mu^2 T^2 + \sigma^2 T = (X_0 + \mu T)^2 + \sigma^2 T$.
- Check: $X_T \sim N(X_0 + \mu T, \sigma^2 T)$, so $\mathbb{E}[X_T^2] = \text{mean}^2 + \text{variance} \checkmark$.

S10.

- $g = e^{x+t}$, so $g_t = g_x = g_{xx} = Y$.
- $dY = (Y + \mu Y + \frac{1}{2} \sigma^2 Y) dt + \sigma Y dW = (1 + \mu + \frac{1}{2} \sigma^2) Y_t dt + \sigma Y_t dW_t$.
- Drift and volatility are proportional to Y : a geometric Brownian motion with drift rate $1 + \mu + \frac{1}{2} \sigma^2$ and volatility σ .

S11.

- (a) $d(W^2 - t) = 2W dW + dt - dt = 2W dW$: martingale.
- (b) $d(W^3) = 3W^2 dW + 3W dt$: drift $\neq 0$, not a martingale.
- (c) $g = w^3 - 3tw$: $g_t = -3w$, $g_x = 3w^2 - 3t$, $g_{xx} = 6w$. Drift $= -3W + \frac{1}{2}(6W) = 0$: martingale.
- (d) $g_t = -\frac{1}{2} \sigma^2 g$, $g_{xx} = \sigma^2 g$: drift $= 0$, martingale (the exponential martingale of S3).

S12.

- (a) $g(t, w) = S_0 e^{(\mu - \frac{1}{2} \sigma^2)t + \sigma w}$: $g_t = (\mu - \frac{1}{2} \sigma^2) S$, $g_x = \sigma S$, $g_{xx} = \sigma^2 S$. Then $dS = [(\mu - \frac{1}{2} \sigma^2) S + \frac{1}{2} \sigma^2 S] dt + \sigma S dW = \mu S dt + \sigma S dW$. ■
- (b) $f = \ln s$, $f' = 1/s$, $f'' = -1/s^2$, with drift μS and volatility σS : $d \ln S = [\mu S \cdot \frac{1}{S} - \frac{1}{2} \sigma^2 S^2 \cdot \frac{1}{S^2}] dt + \sigma S \cdot \frac{1}{S} dW = (\mu - \frac{1}{2} \sigma^2) dt + \sigma dW$.
- (c) $d(W^4) = 4W^3 dW + 6W^2 dt$. Integrate and take $\mathbb{E}$: $\mathbb{E}[W_T^4] = \int_0^T 6\mathbb{E}[W_t^2] dt = \int_0^T 6t dt = 3T^2$.
- (d) Property (1) uses $\mathbb{E}[f(t_{i-1}, W_{t_{i-1}}) \Delta W_i] = \mathbb{E}[f] \mathbb{E}[\Delta W_i] = 0$, which needs f to be independent of ΔW_i . At the right endpoint $f(t_i, W_{t_i})$ is correlated with ΔW_i ; for $f = W$, $\mathbb{E}[W_{t_i} \Delta W_i] = \Delta t$, and the sum of these is $T \neq 0$.

Appendix A. Slide coverage checklist

Slide	Content	Section
S1	Title: Session 4, Stochastic Integrals and Itô's Formula	header
S2	$S_t = S_0 + \mu t + \sigma W_t$; need drift and volatility; $S_t \sim N(S_0 + \mu t, \sigma^2 t)$	1.1
S3	Integral vs differential form; $\mu(t, S_t)$, $\sigma(t, S_t)$; constant-coefficient solution	1.1
S4	Future = present + deterministic + stochastic; question on the drift	1.2
S5	Limit to SDE; BM nowhere differentiable	1.2
S6	S_T as Riemann + stochastic integral; Riemann definition	1.3
S7	Riemann integral as area; 5-interval picture	1.3
S8	Definition of the stochastic integral; left endpoint; previsibility	1.4
S9	Five properties of I_T	1.5
S10	Proof of property (1)	1.6
S11	Proof of property (2)	1.6
S12	Example: mean and variance of $\int_0^T W dW$	1.7
S13	Strong law of large numbers; $(\Delta W)^2 \sim \Delta t$	1.8

Slide	Content	Section
S14	Box calculus rules and table; $(dX)^2 = \sigma^2 dt$	1.8
S15	History: Brown, Einstein, Perrin, Wiener, Itô	1.9
S16	Kiyosi Itô portrait and quote	1.9
S17	Why Itô's formula (theory and practice)	1.10
S18	Quotes (Hamming, Föllmer)	1.10
S19	Ordinary Taylor expansion $\Rightarrow dx/dt = f'(t)$	1.10
S20	Itô for $f(W_t)$: the extra term	1.11
S21	Itô for $f(S_t)$ with $dS = \mu dt + \sigma dW$	1.12
S22	Itô's formula for $f(X_t)$; example $Y = X^2$	1.12
S23	Itô's formula for $g(t, X_t)$; example $Y = e^{X+t}$	1.13

Appendix B. Flag summary (△!)

- [S3] “A stochastic process is a continuous process that can be written” as drift + dW integral: this defines an Itô process, not stochastic processes in general.
- [S4] The drift question is posed but not answered on the slides: the drift is deterministic conditional on time- t information (1.2).
- [S7] Typo: “would like to integral” $\rightarrow$ “integrate”.
- [S8] Summand written $f(t_{i-1})$ for an integrand $f(t, W_t)$; read $f(t_{i-1}, W_{t_{i-1}})$. “Exists in an appropriate sense” = mean-square (L^2) limit.
- [S9] Property 4 has a redundant outer $\mathbb{E}$ for deterministic f, g ; properties 2 and 5 are the same statement; property 3 needs a deterministic integrand ($\int W dW$ is not normal).
- [S10] $\mathbb{E}$ and $\lim$ swapped without justification; summation index typo “ $i = i$ ” $\rightarrow$ “ $i = 1$ ”.
- [S13] SLLN written as $P(\bar{X}_n = \mu) = 1$ for large n ; correct form is $P(\lim_{n \rightarrow \infty} \bar{X}_n = \mu) = 1$.
- [S10–S11] Notation: $\Delta W_{t_{i-1}}$ denotes the forward increment $W_{t_i} - W_{t_{i-1}}$.
- [S12, S22, S23] Examples are posed without solutions on the slides; the solutions here (1.7, 1.12, 1.13) are my own and should be checked against the lecture.